

FINAL PROJECT PRESENTATION

Voice-Based Speaker Verification

Secure user authentication under spoofing and
noise

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01

SECTION 01

The problem

Why a voice alone is not enough to prove who you are.



THE PROBLEM

Why voice authentication is hard

Voice is the most convenient credential we have. It is also the easiest one to record, and the first to break when the room gets loud.



RISK 01

Spoof and replay

An attacker plays back a recording of the real user.

RISK 02

Background noise

Street, office and babble noise break the match.

DEFINITIONS

What we measure

Equal Error Rate, or EER, is the operating point where false accepts and false rejects are equal. Lower is better.



SV

Is this the claimed speaker?

Compares a probe against one enrolled template.

ANTI-SPOOF

Is this live speech or a fake?

Flags replayed and synthetic audio before scoring.

SASV

Both checks in a single score

Accepts only a genuine speaker speaking live.

Protocol discipline

Thresholds are tuned on the development set. The evaluation set is reported once, with no further tuning.

Why this matters

Four reasons this is worth building, and worth trusting.



Frictionless

No password typing and no code to wait for on repeat logins.



Spoof aware

A recorded voice is detected and rejected before any scoring.



Noise ready

Enhancement and learned fusion hold accuracy up in noise.



Deployed

A running web stack with automated build and deploy to cloud.

Fintech · call centres · access control · passwordless consumer apps

02

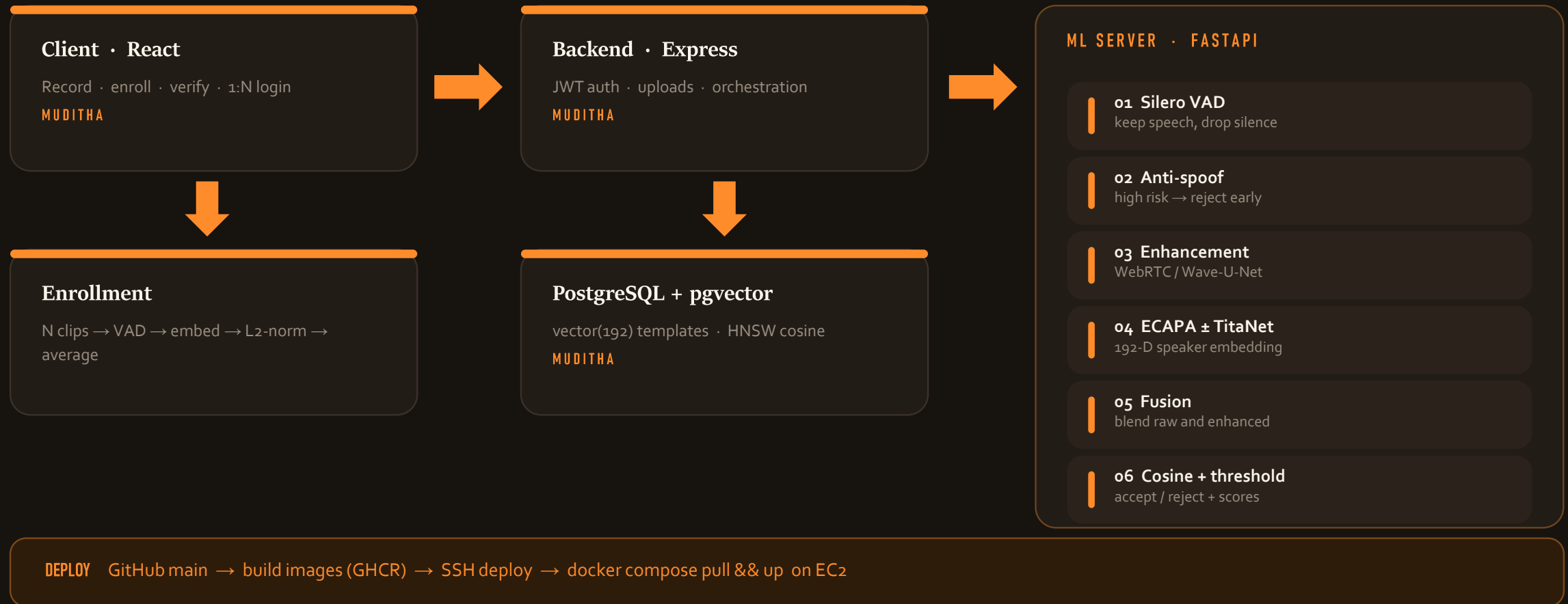
SECTION 02

The system

One architecture from the browser to the speaker model.

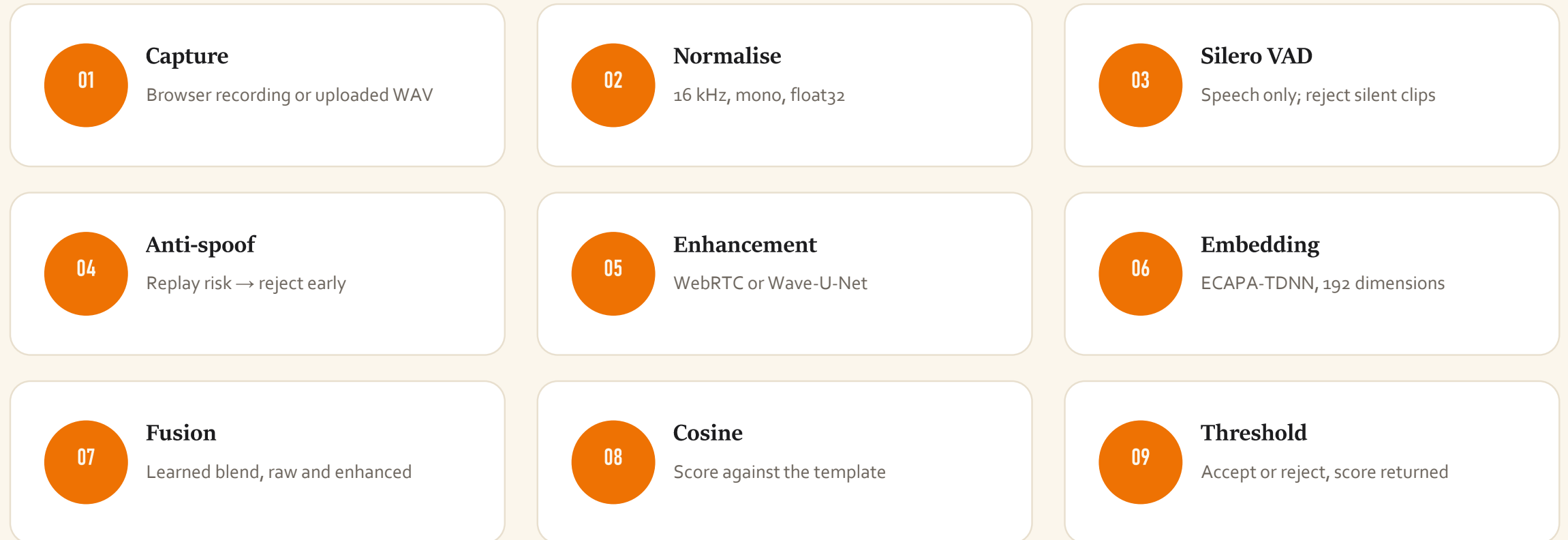


End-to-end architecture



The verification pipeline

Nine steps from a microphone to an accept or reject decision.



ENROLLMENT several utterances → VAD → embed → L2-normalise → average → store in pgvector

Technology stack

A conventional web stack in front of a speech model server.

Application layer	Technology
Frontend	React + Vite, MediaRecorder
Backend	Node.js / Express, JWT, Multer
Database	PostgreSQL + pgvector, HNSW
API surface	/enroll/template, /verify, /replay
Deployment	Docker Compose, GHCR, Actions

Model layer	Technology
Speech gating	Silero VAD (ONNX)
Speaker encoder	SpeechBrain ECAPA-TDNN
Second encoder	NVIDIA NeMo TitaNet-Large
Enhancement	WebRTC NS, tuned Wave-U-Net
Anti-spoof	Replay CNN, AASIST, LFCC

One contract across the whole system

Audio is always 16 kHz mono float32 and embeddings are always 192-D and L2-normalised, so the product path and the research notebooks score the same way.

Key design decisions

Six choices that shaped the system, and the reason behind each.

Average several enrollment clips

A single utterance overfits to one recording session.

Run VAD before scoring

Silence and room tone corrupt the embedding.

Never enhance blindly

Enhancement can hurt, so the fusion learns when to trust it.

Put the spoof gate inside the decision

Security first: reject replay before identity matching.

Two encoder slots in the database

ECAPA and TitaNet templates live side by side.

Compose deploy instead of k3s

Fits a small, low-memory cloud instance.

03

SECTION 03

The business

Who pays for this, and what they measure it by.



Where this creates value

The same pipeline serves a bank, a call centre and a door. What changes between them is only the risk threshold.



WHO BUYS IT

- Banks and fintech apps
- Call-centre identity
- Building and device access
- Passwordless consumer apps

WHAT THEY GET

- Fewer steps per login
- Replay-attack resistance
- Works in noisy channels
- On-premise or cloud

HOW WE SHIP IT

- Enroll, verify, identify API
- Threshold and risk dashboard
- Pilot on one workflow
- Tuned per customer

The numbers a customer asks for

Every claim below comes from a run in this repository,
including the ones that did not go our way.

KPI	Why the business cares
EER / FAR / FRR	Security versus user friction
Spoof catch rate	Direct fraud prevention
Verification latency	Perceived speed of the login
Cost per verification	Unit economics at scale
Enrollment time	Onboarding drop-off

EVIDENCE WE ALREADY HAVE

0.83%

SASV-EER, clean

8.79%

all-noisy SV-EER

0.54%

VoxCeleb EER

Live

cloud deployment

Responsible use

Biometric consent and a retention policy are mandatory, and high-risk transactions should still use a second factor.

04

SECTION 04

The research

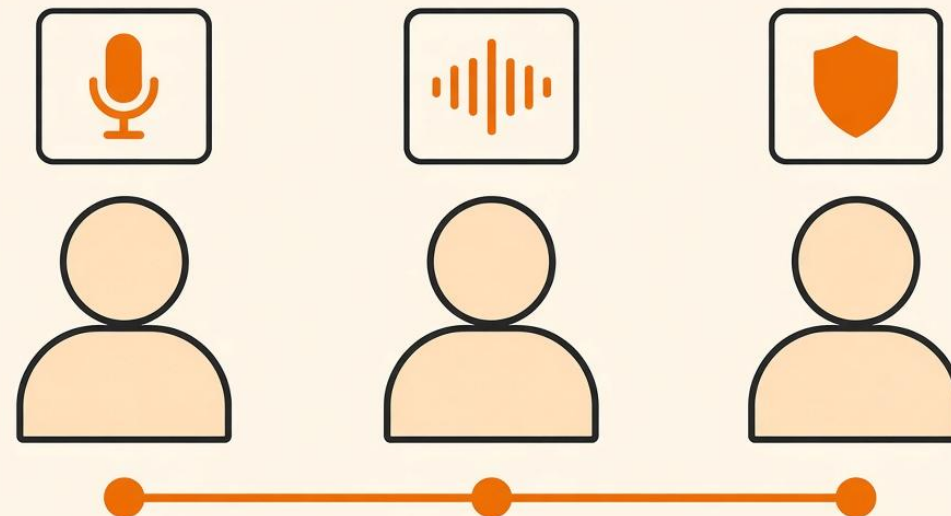
Three tracks, three protocols, and one result we did not want.



Who built what

Identity, robustness and security were developed and measured separately, then integrated into one pipeline.

The cards on the right are clickable.



Muditha Herath

Speaker models, Silero VAD, pgvector and the full app and ML integration

Yasith Hewarathna

Noise resilience, enhancer comparison, Wave-U-Net and learned fusion

Herath H.M.M.P.B

Replay and spoof detection, SASV protocol, noise-gated SASV and CI/CD

Speaker core and product

The part of the project that actually runs in production: capture the voice, gate it, turn it into a vector, and search it.



- Integrated ECAPA-TDNN and NVIDIA TitaNet-Large into the ML server, with offline checkpoint loading
- Added Silero VAD to the live inference path, including zero-speech fallback handling
- Migrated storage to PostgreSQL + pgvector with HNSW cosine indexes for 1:1 and 1:N search
- Built the Express API, the React recorder and the verification result view
- Authored the VAD, audio and schema test suites with pytest markers

Why a dual-encoder design

ECAPA is small, open and fast on CPU, which suits the deployed service. TitaNet is kept as a second template slot so a stronger encoder can be swapped in later without a schema change.

Does removing silence help?

VoxCeleb1 test speakers, 1,120 same-speaker and 1,120 different-speaker pairs over 320 files.

0.54%

EER WITH SILERO VAD

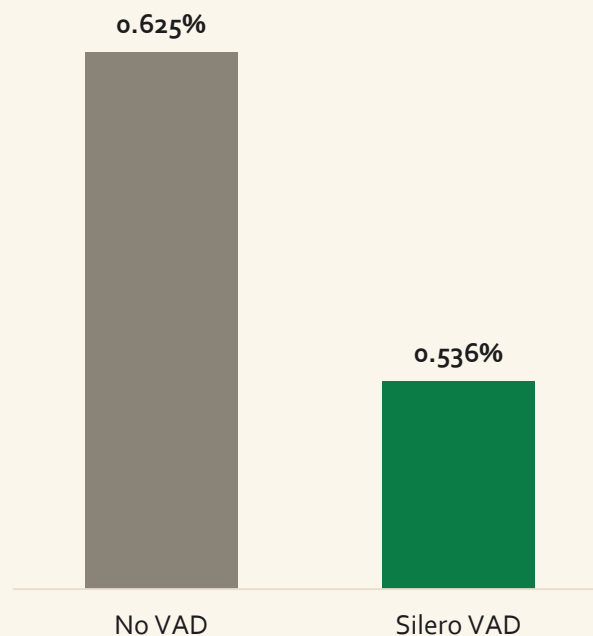
down from 0.625% on raw audio, a gain of 0.09 points

92%

speech kept

320

trial files



Reading the result honestly

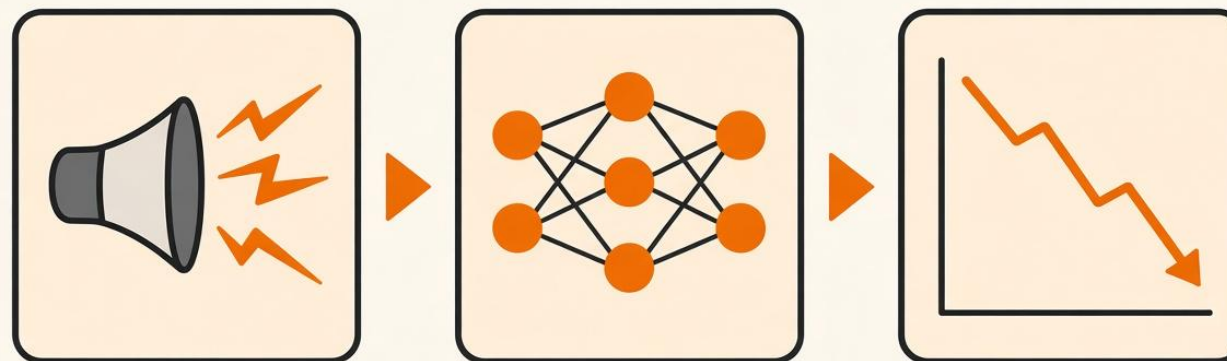
The EER gain is small, because VoxCeleb clips are already almost all speech.

The bigger practical win is robustness: silent or empty recordings are now caught instead of producing a meaningless embedding.

The TitaNet comparison runs on the same trial list and is blocked only on the NeMo runtime install.

Noise-robust verification

Babble is the hardest condition in the whole study, and generic enhancement often makes verification worse rather than better.



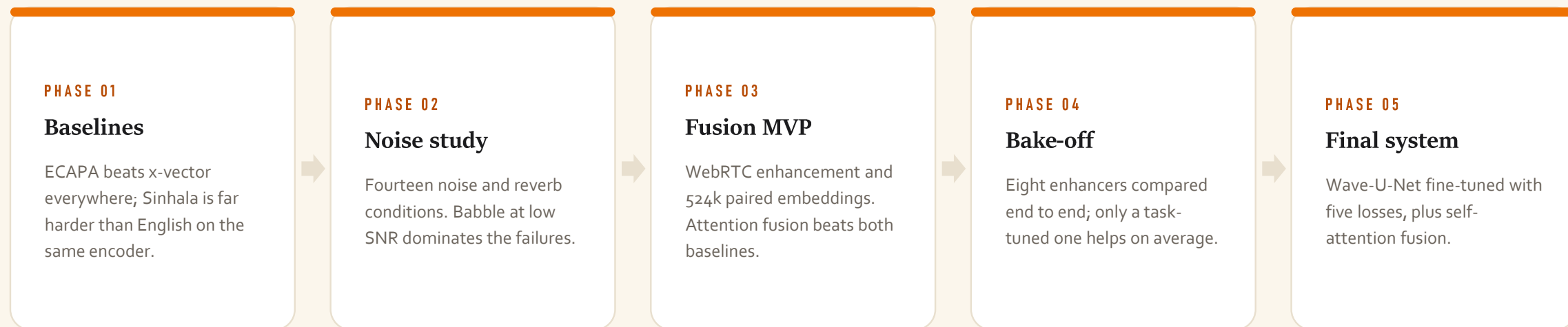
The idea

Do not choose between raw and enhanced audio. Embed both with ECAPA and let a network learn how to blend them per utterance.

The final system

A Wave-U-Net enhancer fine-tuned with five speaker-aware losses, followed by self-attention fusion.

From a fast baseline to a tuned enhancer



The negative result that shaped the design

DeepFilterNet3 enhancement helped only on babble at moderate SNR and hurt on clean, reverberant and stationary-noise audio. That is exactly why the final system learns a blend instead of always enhancing.

Final results on VoxCeleb1 + MUSAN

37,611 trials, three noise types across nine SNR levels plus clean. EER in percent, lower is better.

8.79%

ALL-NOISY EER AFTER FUSION

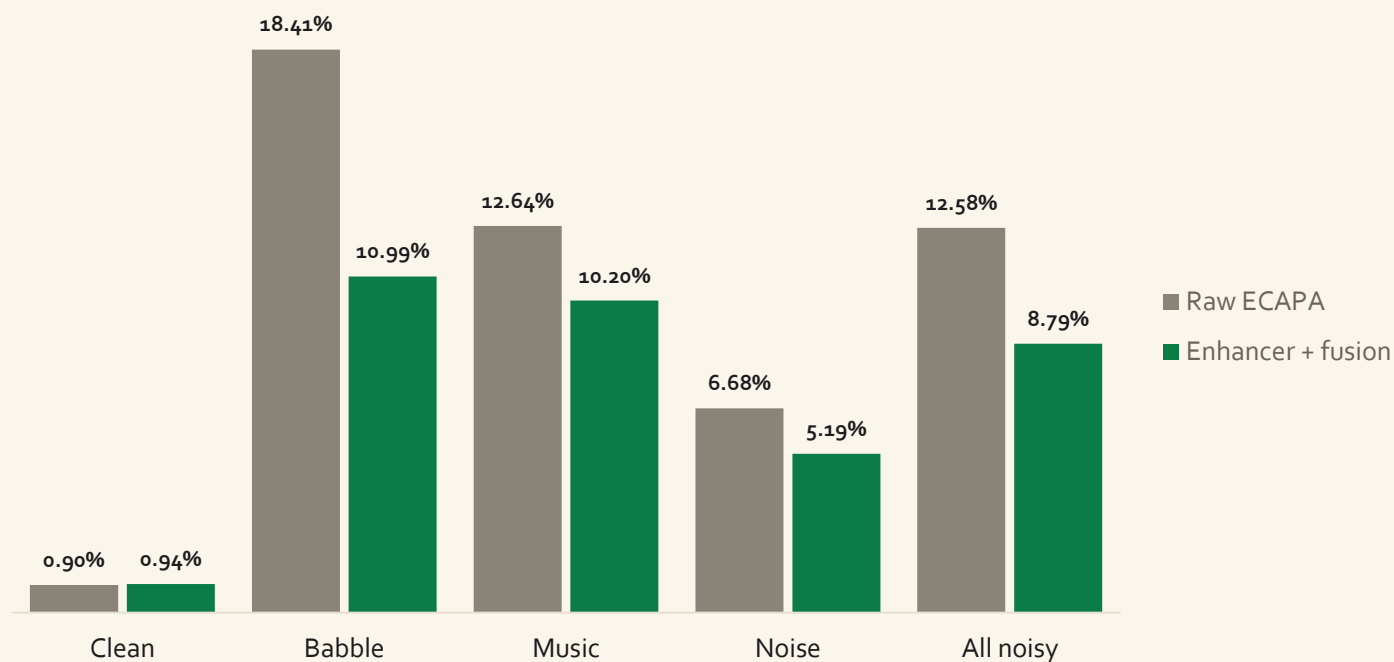
down from 12.58% on raw ECAPA, a gain of 3.78 points

+0.03

clean EER cost

14.62%

hardest subset



The fused system wins at every SNR and every noise type, with no crossover, and gives up almost nothing on clean audio.

Detecting a replayed recording

ASVspoof 2017 V2 physical replay attacks, 1,040 held-out development files. Same CNN throughout; only the front-end changes.

4.90%

EER WITH AN INVERTED-MEL FRONT-END

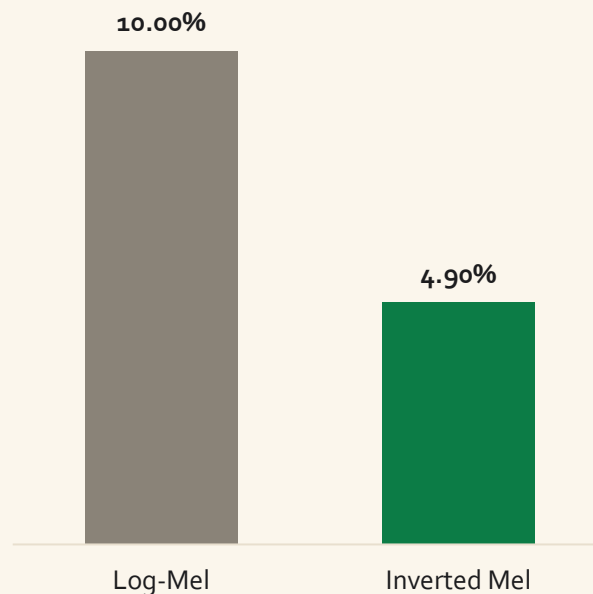
down from 10.00% with a standard Log-Mel front-end on the identical network

0.955

replay F1

95.1%

accuracy



Why the front-end decides this

A Mel filterbank spends most of its resolution on low frequencies, where the speech energy is.

Inverted Mel spends it on high frequencies instead, which is exactly where a playback loudspeaker and a second microphone leave their fingerprint.

The network, the data and the training schedule were unchanged, so the gain is attributable to the front-end alone.

Where replay detection stops working

Every row is a zero-shot run: trained on one corpus, scored on another with no retraining.

Trained on → scored on	EER
ASVspoof 2017 → 2019 PA	50.47%
2019 PA → ASVspoof 2017	39.24%
2017 + PA → 2019 LA (TTS/VC)	41.12%
AASIST off-the-shelf → 2017	64.32%

MIXED-CORPUS TRAINING, THREE FRONT-ENDS

Front-end	2017	PA 2019
Log-Mel	21.17%	5.99%
Inverted Mel	10.12%	9.66%
LFCC	9.24%	8.98%

Fix 1 — train on the target corpus

Training the same network directly on ASVspoof 2019 PA reaches 7.96% EER over 15,434 development files, at an F1 of 0.949.

Fix 2 — mix the corpora

One corpus-balanced model holds on both instead of collapsing on the unseen one: 9.18% on 2017 and 11.00% on PA.

The consequence — why AASIST was needed

At 41.12% on logical access, a replay detector is blind to synthesised and converted speech. A different countermeasure has to carry that attack, which is the next slide.

Anti-spoofing and the joint SASV metric

A recording of the real user passes speaker verification.
That is the hole the countermeasure has to close.

- Evaluated on ASVspoof 2019 LA with the official SASV 2022 trial lists
- MUSAN noise added on the test side only, at clean / 15 / 10 / 5 / 0 dB
- Tuned on dev, locked the evaluation set once, with no further tuning

System	What it combines
Bo	ECAPA speaker score only
B1	ECAPA + AASIST countermeasure ($\alpha = 0.30$)
P2 / P3	SNR-gated and always-on enhancement

SV-EER

target vs non-target

SPF-EER

target vs spoof

SASV-EER

both at once

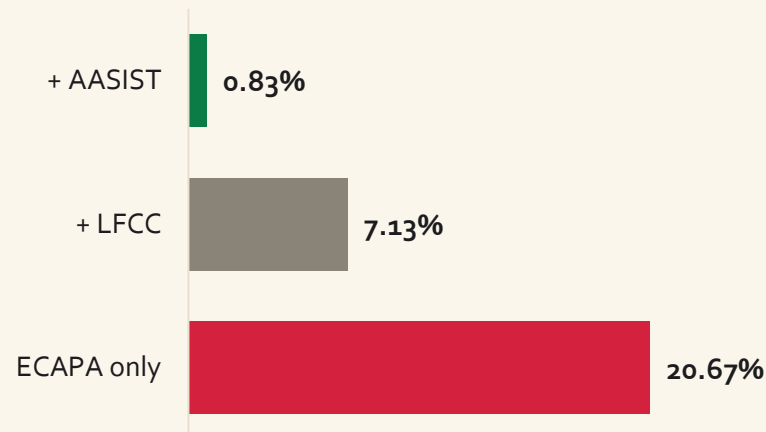
One clear win, one honest failure

Locked evaluation on clean audio, then the same systems under test-side noise.

0.83%

SASV-EER WITH AASIST FUSION

down from 20.67% for ECAPA alone, and better than the published 1.71% baseline



System @ 15 dB	SASV-EER
Bo raw ECAPA	16.91%
P2 gated enhance	16.93%
P3 always enhance	23.11%

Result 1 — the countermeasure is essential

A strong speaker encoder on its own fails against spoofed speech, at 27% spoof error. Fusing AASIST with a tuned weight closes it.

Result 2 — an honest negative

Gating the enhancer by estimated SNR did not beat raw ECAPA on SASV, and always enhancing was clearly worse.

How the three tracks fit together

IDENTITY

Is this the same speaker?

ECAPA and TitaNet embeddings, VAD gating and a pgvector template store.

MUDITHA

ROBUSTNESS

Can we still verify in noise?

A task-tuned Wave-U-Net plus learned fusion of raw and enhanced embeddings.

YASITH

SECURITY

Is the audio real?

A replay CNN and the AASIST countermeasure fused into a joint SASV decision.

H.M.M.P.B

Reconciling the two enhancement findings

Learned fusion improved speaker verification under MUSAN noise on VoxCeleb1, while a simple SNR-gated enhancer failed on the SASV task on ASVspoof LA. These are different protocols, different attacks and different decision rules, so both results stand: enhancement pays off only when the blend is learned for the exact task.

DEMONSTRATION

The system in use

Enroll once, verify in a second, and watch a replayed recording get turned away.

If the room audio is bad

Play the recorded walkthrough, then show the green deployment run and the live service.



- Sign in, then enroll with a few short recordings
- Verify with a fresh recording and read the score
- Replay a recording into the mic and watch it be rejected
- Push to main and see the cloud stack rebuild itself

Limitations and next steps

WHAT WE WOULD NOT CLAIM

EER values come from different protocols and are not directly comparable across tracks

Noise was applied mainly on the test side while enrollment stayed clean

Replay detection does not transfer across corpora, so new microphones and rooms need their own data

Parts of the noisy-SASV grid are smoke-scale rather than full runs

WHAT WE WOULD DO NEXT

Apply the learned fusion to the SASV task instead of a fixed SNR gate

Complete the countermeasure-under-noise matrix and lock it once

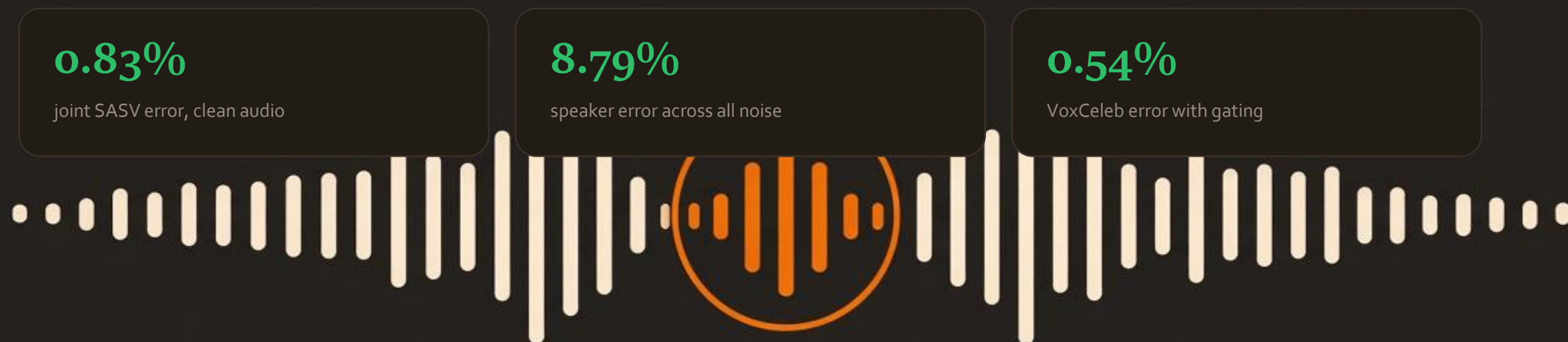
Calibrate thresholds per SNR band so the decision stays stable

Finish the TitaNet comparison and collect real device recordings

REPORTING DISCIPLINE every number in this deck comes from a run we can reproduce, including the experiments that did not work.

IN SUMMARY

Designed, measured, and deployed



We built the running product, measured every track on a standard protocol, reported the experiments that failed, and automated the deployment.

Thank you — questions?

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